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National University of Computer and Emerging Sciences, Lahore Campus



Course: Computer Organization and Assembly Language
Program: BS(Computer Science)
Duration: 15 Minutes
Paper Date: 12th Sept, 2018
Section: B
Exam: Quiz-1

Course Code: EE213
Semester: Fall 2017
Total Marks: 10
Weight:
Page(s):

Q1. Consider the following listing file below. This program is currently in execution. Current value of CS Register is 0x0A10 and the Physical address pointed to by CS:IP is 0A20C. For this program, determine what will be the **current contents** of a) IP register, and b) ax register, given the above CS:IP values. **[4 Marks]**

1	[org 0x0100]
2 00000000 B80A00	MOV AX, 10
3 00000003 B80600	MOV AX, 6
4 00000006 B80500	MOV AX, 5
5 00000009 B80200	MOV AX, 2
6 0000000C B80100	MOV AX, 1
7 0000000F B8004C	mov ax, 0x4c00
8 00000012 CD21	INT 21h

IP 010C

AX 2

Q2. Calculate the physical address generated by the following and identify if there is a carryout. **[2 Marks]**

Logical Address	Physical Address	Carry out?
FFFF:4006	1 03FF6	Yes
AB01:FFFF	BB00F	No

Q3. My processor has a RAM of 2 MB. How many address lines do I need on my address bus? **[1 Mark]**

Sol: 21

Q4 What will be the values of following after execution of each operation? Each is an independent statement. **[1 Marks]**

		CF	OF
1.	mov ax, 0x0120 xor ax, 0x1102	0	0

Q5. What will be the flags values after the following code: **[2 Marks]**

```
[org 0x0100]
mov al, -1
add al, 2
sub al, 2
mov ax, 0x4c00
INT 21h
```

AF	1
ZF	0
OF	0
CF	1